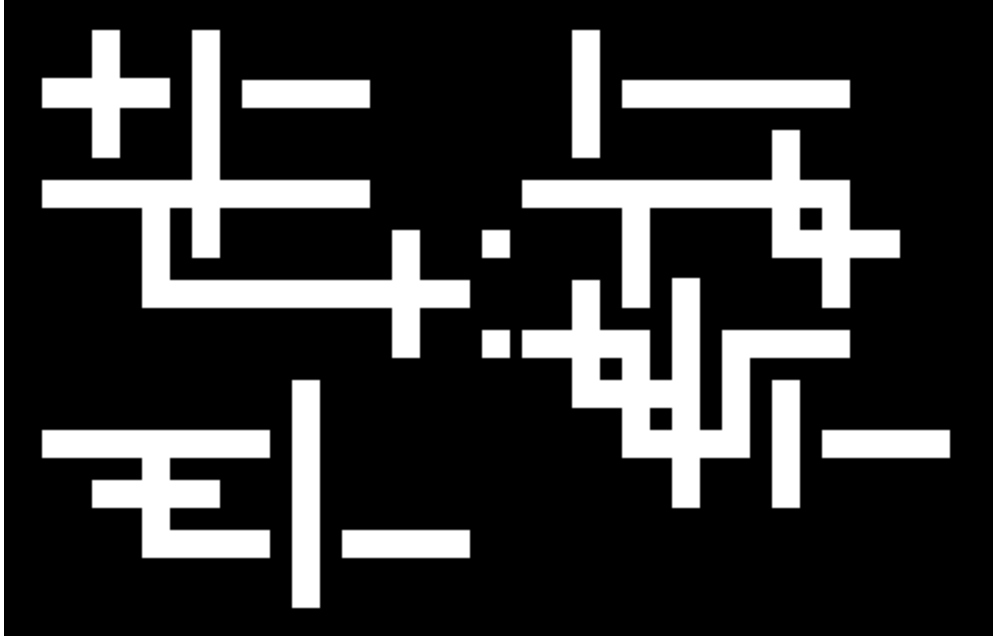


StarLoop: Crisis

DESIGN DOCUMENT

By Sirius)(Black for the 2024 Pirate Software Game Jam

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Introduction

Game Summary Pitch

Starloop is an escape room, but instead of escaping a room, you're trying to stop your spaceship from crashing into the sun, as an unknown pathogen is eating the wiring of your spaceship

Inspiration

The witness/Tunic

A pair of those life-changing games one dreams to re-create, i'm not saying i will, but i'm hoping to emulate some of their design philosophies to let the rules of the world be alien and unexplained, yet discoverable.

Wireworld

Of course, Wireworld is the foundational mechanic, this is how the wiring of the spaceship works, although of course turning it into

3d and with more complex mechanics to allow for actual puzzle creation.

Outer Wilds

A time loop mechanic is always interesting, and with starloop, it could give you the space to explore the problem at your own pace, even though your demise is imminent. but i really enjoy how outer wilds gave an in-universe explanation for why/how the time loop works.

Exodus Burned

The original pitch for this game was something along the lines of “you wake up from your stasis pod, only to find that the spaceship’s AI has gone rogue, steering you into the nearest star”, and it has always bothered me that the game they ended up making was very different from this pitch

Player Experience

In a 3d voxel environment, the walls are the wires, buttons connect with wires to doors, and the player is tasked with finding a way to change the spaceship’s course, before crashing into the rapidly approaching star, as an unknown pathogen slowly consumes the ship, messing with it’s systems, alive, and... intelligent?

Platform

The game is developed to be released on windows PC

Development Software

- Godot 4.2 with C#/Rider for programming
- Aseprite /goxel for graphics and UI

Genre

Singleplayer, puzzle, casual

Target Audience

With a ramping in difficulty, and potentially unintuitive mechanics, this game is marketed to veteran puzzlers that like to chew on a huge puzzle, while also providing a ramping of easier sub-puzzles to learn how the mechanics interact.

Concept

Gameplay overview

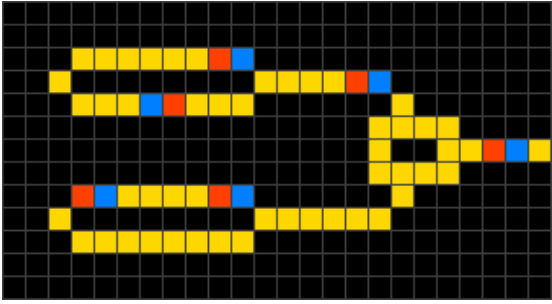
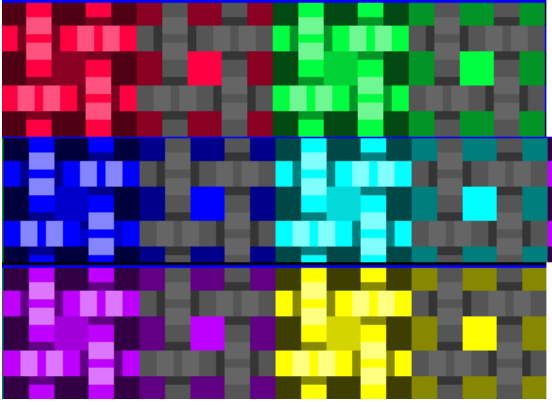
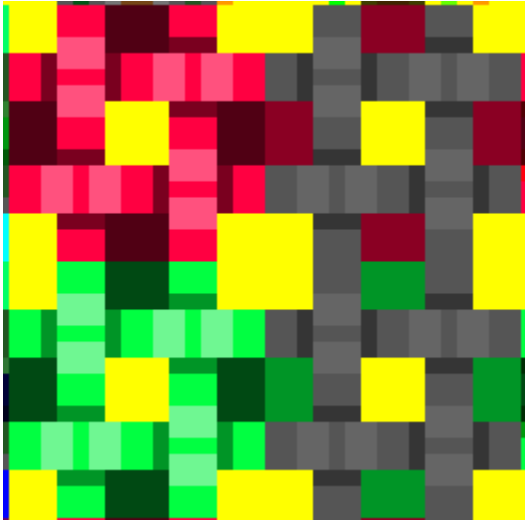
The player explores a spaceship that is slowly deteriorating, as a rapidly spreading pathogen “eats” the ship’s wiring, destroying system after system, as the player learns where the pathogen is, and how the wiring system functions, they can perform repairs, cutting off the infection and keeping the ship functional.

Theme Interpretation (It’s spreading)

[The pathogen slowly spreads throughout the ship, travelling through walls and affecting different systems](#)

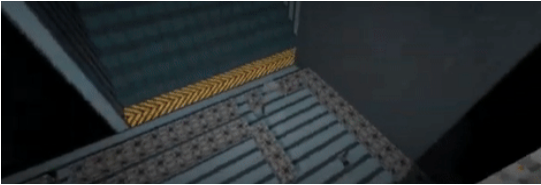
As the pathogen spreads throughout the ship, various systems will begin to fail as their electrical systems break down, and cutting off one food source will make the pathogen spread to other systems faster as it hunts for the resources it needs to spread.

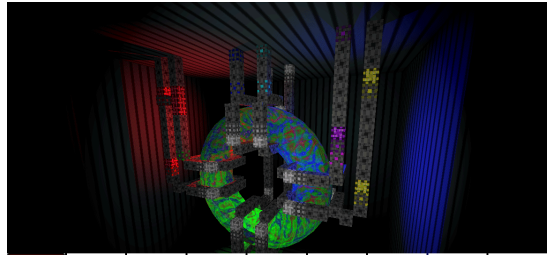
Primary Mechanics

Mechanic	Animated Mockup (Art not necessarily final)
<u>Wireworld</u> The basic mechanic is wireworld	
<u>Color theory</u> Instead of one wire carrying one type of signal, signals can be colored	
<u>Filter cells</u> Certain cells can only accept signals of matching colors	

Mechanic	Animated Mockup (Art not necessarily final)
<u>Virus</u> A virus is slowly spreading through the wiring, altering it's behavior and disrupting the ship's functionality.	

Secondary Mechanics

Mechanic	Animated Mockup (Art not necessarily final)
<u>Time loop</u> The ship's emergency systems have trapped you in a time loop, allowing you to try different strategies to out-smart the infection	

Mechanic	Animated Mockup (Art not necessarily final)
<u>Everything is a wire</u> Wires control everything, doors, lighting, and all of the ship's systems, and because they move quickly, it may be difficult to see the current flowing without specialized equipment	 0 0 0 0 0 0 0 0 0 0

Art

Design

A very **minimalistic** approach will go into the design of the game, relying fairly simple pixel art combined with voxel meshes to do most of the heavy lifting, with a small nod to the old days of unreal for the skybox

Audio

Music

If only i had time, i just took something from an asset pack i had laying around

Sound Effects

Same here, very much don't have time for this, if anything i might add a small sparking sound when a voxel fails to trigger like it's supposed to, though that has a fairly high chance of happening constantly

Game Experience

UI

I mean... there's a toolbar, And an epilepsy warning due to the flashing and a main menu, not much more thought went into it, again, due to time constraints.

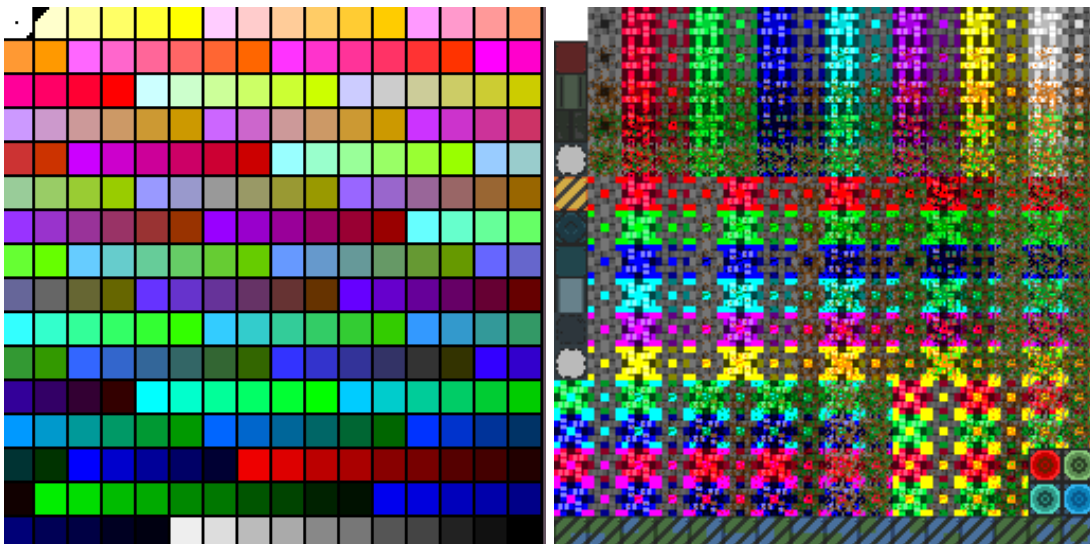
Controls

Keyboard

WSAD - move
 shift to sprint,
 space to jump,
 left mouse to place,
 right mouse to remove,
 Middle mouse button to pick the selected block
 Scroll to pick a different block
 1-9 to pick hotbar items (sorry, they don't persist between loops)

Save format:

Chunks are loaded from the .gox files in the Voxels folder, and tiles from the sprite sheet map directly onto the MagicaVoxel palette



Development Timeline

MINIMUM VIABLE PRODUCT

#	Assignment	Type	Status	Finish By	Notes
1	Design Document	Other ▾	Finished ▾	Jan 16, 2...	Yee try 26th
2	Player movement	Coding ▾	Finished ▾	Jan 12, 2...	Walking mechanics
3	Render Voxels	Coding ▾	Finished ▾	Jan 12, 2...	Draw voxels, needs custom mesh building
4	Pick voxels	Coding ▾	Finished ▾	Jan 14, 2...	Need to pick chunk and index based on raycast
5	Edit voxels	Coding ▾	Finished ▾	Jan 14, 2...	First single-voxel edits, saving/loading
6	Fast edit voxels	Coding ▾	Finished ▾	Jan 17, 2...	Fill lines, planes and boxes in order to more rapidly edit, paint cells, also limit max range to prevent accidental edits
7	Wire world	Coding ▾	Finished ▾	Jan 20, 2...	Quick updates to cell states for UV changes, basic RGB/3d wireworld
8	Infection	Coding ▾	Finished ▾	Jan 22, 2...	Secondary cellular automaton
9	Wireworld connect to objects	Coding ▾	Finished ▾	Jan 25, 2...	elevators/doors/etc that require being powered by wiring

#	Assignment	Type	Status	Finish By	Notes
10	Player controls	Coding ▾	In progress ▾	18 Jan 2...	Player should probably not have full world edit abilities Still no clue what they should have, though...
11	Level design	Art ▾	In progress ▾	Jan 26, 2...	Design the layout/puzzles of the ship Endless torture...
12	Main menu	Art ▾	crammed... ▾	Jan 26, 2...	Opening screen
13	Sound effects	Audio ▾	Not started ▾	Jan 26, 2...	Doors opening, footsteps, elevators,
14	music	Audio ▾	crammed... ▾	Jan 26, 2...	Environmental music/sounds
15	Pause menu	Art ▾	Not started ▾	Jan 26, 2...	Better than just showing the cursor
16	Any extra polish	Other ▾	Not started ▾	Jan 27, 2...	
17	SUBMIT	Other ▾	In progress ▾	Jan 27, 2...	Create Itch Page and upload

BEYOND (if ahead of schedule / extra time)

Time loop	Coding ▾	Finished ▾	Death mechanic, visually reverse actions,
Bigger ship	Art ▾	Not sta... ▾	Make ship look better and make it larger
"Stroboscopic" signals	Coding ▾	Finished ▾	Wire signals are only visible in designated areas.

Settings	Coding ▾	Not sta... ▾	👤 they exist in games
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